

TOP TECH TRENDS OF 2011



Everyone has his own opinion when it comes to compiling

tech trends but the line of thought essentially remains more or less the same. Things that created a buzz in the industry to things that fell flat on their face, here's an informative info-graph summing it up: bit.ly/t2zLI1

FUEL CELLED APPLE PRODUCTS

Recent patent applications filed by Apple indicate that fuel celled products might be in the making. It's always hit idea with consumers when it comes to saving the energy and Apple seems to be moving ahead of its competitors again. Soon we might have iMacs powered by fuel cells: bit.ly/sb4RGI

MOST INNOVATIVE COUNTRIES



Here's an info-graph showcasing the most innovative countries and industries on the planet. US and Japan are obviously there, but China is absent! bit.ly/sreSQy

ANATOMY OF A SKYSCRAPER

Modern architecture has blessed us with many things, whether it's the sheer size or magnanimity of a structure or its purpose, modern day skyscrapers are the landmarks of the present day world. One such tall tower is scraping its way up the skies of Europe and once completed it will be the tallest tower in the continent. Here's more on the story: bit.ly/vXpPxu

should fare a bit better; Intel will launch LGA2011-compatible eight-core processors under the Xeon E5 family sometime next year.

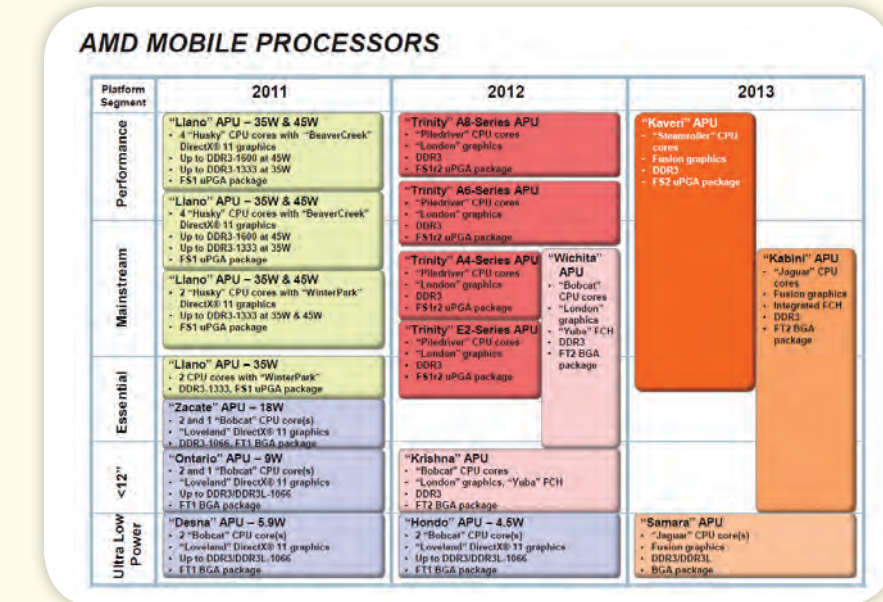
AMD: Mostly mobile, talking tablets

The cloud of uncertainty surrounding AMD and the recent foundry debacle makes it harder to predict what the company will do in 2012. AMD won't unveil its strategy for the year until February, where its CEO, Rory Read, will undoubtedly leverage his expertise in sales to put on a good show. Tablets will likely be a prominent feature, though the 28nm delay means that any parts AMD ships in this segment for 2012 will likely be demo units as opposed to commercially viable products.

The reality underneath the hype is probably fairly grim. AMD will launch Trinity early in Q1, possibly as early as CES. That chip pairs a second-generation Bulldozer core (Piledriver) with a GPU based on the HD 6000 series. Early performance suggests the GPU could be 20-25 percent faster than Llano's. The interconnect between the two components has also reportedly been improved.

On the CPU side of the equation, however, Piledriver will likely struggle to match Llano's performance and power consumption. AMD leaned heavily on Llano's GPU performance and media capabilities when it came to marketing the chip, and it'll likely do the same with Trinity. Tweaked Brazos products, still on 40nm, will round out most of 2012.

Read will almost certainly announce a new 28nm family of Brazos chips built at TSMC, probably with a late-2012 launch date. One of his major tasks, come February, is to sell investors and OEMs on the idea that AMD can still compete with comparatively low CPU performance on the one hand and a slow move to 28nm on the other. With Intel focusing on smartphones and relatively high-end laptops,



AMD should be able to maintain a grip on the mobile market and keep itself in the black.

Graphics presents an interesting situation. The good news is that AMD's next-generation, 28nm Radeon 7000 series — which launched today — will beat Nvidia's 28nm "Kepler" GPU to market by several months. Current rumors suggest that Kepler's high-end version could slip to the latter half of 2012, giving AMD a huge lead to market.

The bad news is that AMD's earnings from graphics sales are pretty terrible. Sunnyvale paid \$5.6 billion to acquire ATI in 2006, then wrote off half that amount (essentially admitting that it paid \$2.4B more than it should've for the Canadian company). The chart above lists AMD's recognized graphics revenue and operating profit/loss for each fiscal year. Note that neither APU sales nor the R&D associated with the development of Brazos, Llano, and Trinity are included.

In the five years since it acquired ATI, AMD has made a grand total of \$190 mil-

lion dollars on \$6.3 billion in sales, for a net profit margin of 3%. This chart doesn't capture the full value of that purchase; the ATI acquisition enabled the development of AMD's current APU and contributed to the success of earlier integrated chipsets. Nevertheless, the data suggests that AMD's ability to leverage graphics earnings to keep the company in the black is extremely limited.

Watch for Sunnyvale to emphasize its longer-term commitments to tablets and the cloud while shipping low-end desktops and laptops. Graphics will be a major focus in all areas; the HD 7000 series could grab market share from Nvidia if it delivers significant performance improvements. Intel's bifurcated strategy on ultrabooks at one end and smartphones on the other creates an area of the market where AMD can effectively market its own products, and Read will try to take full advantage of it.

[Image credit: Sonic840]